

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I Pawan Kumar C/19251146 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Pawan Kumar C

Level - 4

BP

1. Formulation of the 8-puzzle as a search problem.

The 8-puzzle consists of a 3×3 board containing eight numbered tiles (1-8) and one blank space. The objective is to move the tiles until they match the desired goal arrangement.

a) State Representation.

A State is represented as the arrangement of the eight numbered tiles and one blank space on the 3×3 grid.

Example:

Initial State:

1	2	3
4	-	6
7	5	8

b) Initial State.

The initial state is the starting configuration of the puzzle. It can be any valid arrangement of the tiles.

Example:

2	8	3
1	6	4
7	-	5

c) Goal Test.

The goal test checks whether the current state matches the desired goal configuration.

Goal state:

1	2	3
4	5	6
7	8	-

If all tiles are in the correct positions and the blank is in the last cell, the goal is reached.

d) Possible Actions.

The blank space can be moved.

- * UP
- * Down
- * Left
- * Right.

A move is allowed only if it stays within the board boundaries.

Why is the Branching Factor at most 4?

The branching factor is the maximum number of successor states from any state.

* A blank tile in the center has 4 possible moves (up, down, left, right).

* A blank tile on an edge has 3 possible moves.

* A blank tile in a corner has 2 possible moves.

Therefore, the maximum branching factor is 4.

2. AI Agent for Hospital patient Triage.

Performance Measure.

The AI agent should be evaluated using the following Performance measures:

- * Correctly identify critical patients.
- * Minimize waiting time for emergency cases.
- * Reduces incorrect patient prioritization.
- * Improve overall treatment efficiency.
- * Maximize patient safety and satisfaction.

Rationality.

A rationality agent selects the action that maximizes the expected performance measure based on the available information.

For a hospital triage agent, an action that considered rational when it:

- * uses the patient's symptoms, vital signs, and medical history.
- * Correctly estimates the severity of the condition.
- * Assign the appropriate priority level.
- * update its decision when new patient information becomes available.

Conditions for Rational Behavior.

The agent acts rationally when:

- * It has accurate and sufficient patient data.
- * It uses reliable medical knowledge or trained AI models.
- * It considers uncertainty in symptoms.
- * It aims to maximise patient safety and treatment efficiency.
- * It adapts its decisions based on new observations.

3) AI Agent for warehouse Robots.

Most Appropriate Agent Architecture:-

Learning Agent.

A learning agent is the most suitable architecture for warehouse robots.

Justification.

warehouse environments are dynamic because:-

- * Obstacles may appear unexpectedly.
- * Inventory locations can change.
- * Robot traffic varies throughout the day.
- * Picking station workloads change continuously.

A Learning agent can:

- * learn the shortest and fastest routes.
- * Improve navigation from experience.
- * Avoid collisions with workers and other robots.
- * Adapt to changing warehouse layouts.
- * Optimize task scheduling and battery usage.

Why other Architectures are less suitable.

- * Simple Reflex Agent :- Responds only to the current situation and cannot handle complex or changing environments.
- * Model-Based Agent :- Maintains an internal model but has limited ability to improve from experience.
- * ~~Goal-Based Agent :- maintains an internal model but has limited ability to improve from experience.~~
- * Goal-Based Agent :- can reach destinations but may not optimize efficiency.
- * Utility-Based Agent :- Optimizes factors such as time and energy but does not automatically improve over time.

Advantages of a Learning Agent.

- * Continuously improves performance.
- * Adapts to environmental changes.
- * Reduces delivery time.
- * Increases warehouse productivity.
- * Minimise errors and collisions.
- * Optimises energy consumption and route planning.

Conclusion:

A Learning Agent is the best choice because it can learn from experience, adapt to changing warehouse conditions, optimise routes, and continuously improve efficiency, making it ideal for modern automated warehouse systems.